Cn lab exam prep

This is When U Wanna

```
enable
configure terminal

! 1. Configure the interface facing the LAN WITH the DHCP Server (e.g., L512)
interface GigabitEthernet0/0
ip address 192.168.12.1 255.255.255.0
no shutdown
exit

! 2. Configure the interface facing the REMOTE LAN (e.g., L509) and add the helper address
interface GigabitEthernet0/1
ip address 192.168.9.1 255.255.255.0
ip helper-address 192.168.12.2
no shutdown
exit

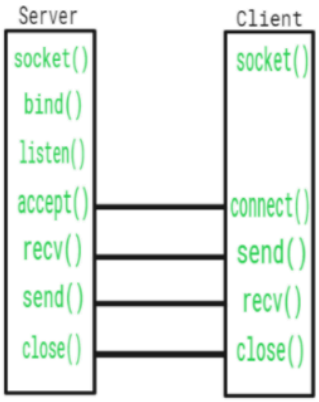
end
```

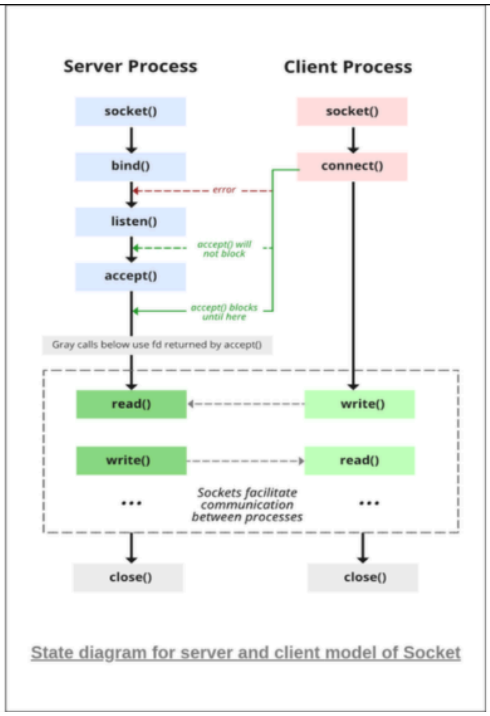
FOR 192.168.1.254 AS DEFULT

```
Router#enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip dhcp excluded-address 192.168.1.1 192.168.1.10
Router(config)#ip dhcp pool LAN1
Router(dhcp-config)#network 192.168.1.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.1.254
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
```

FOR 192.168.2.254 AS DEFAULT

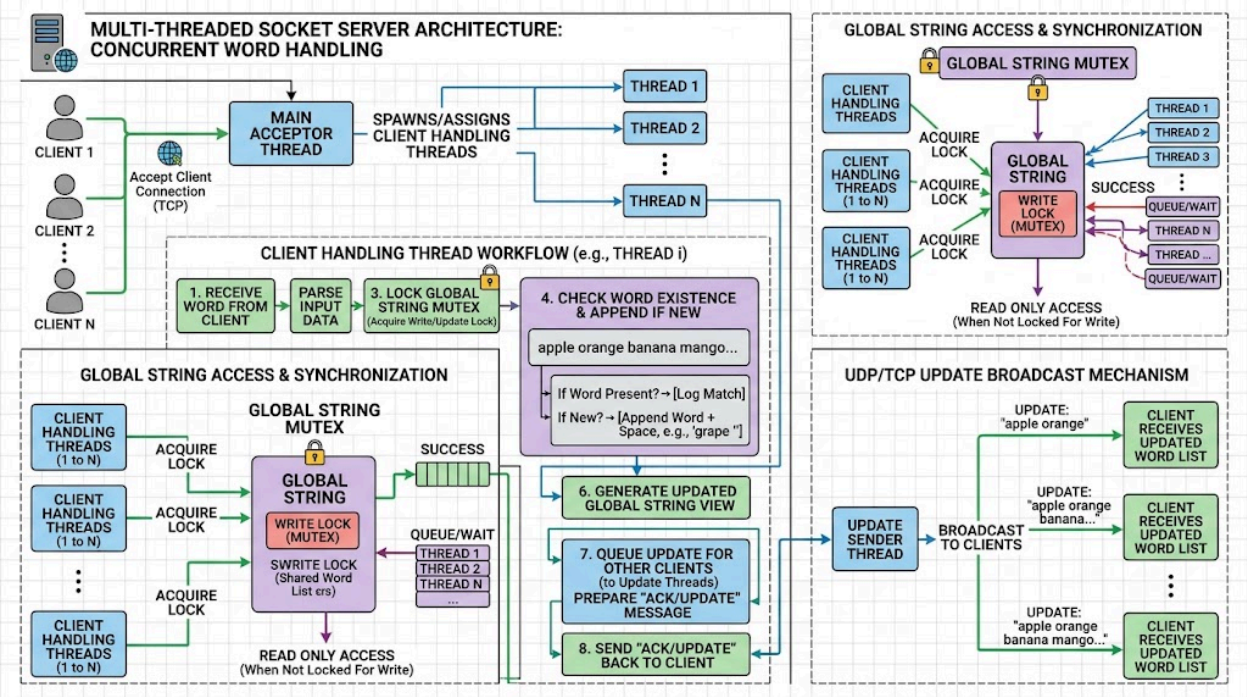
```
Router#enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip dhcp excluded-address 192.168.2.1 192.168.2.12
Router(config)#ip dhcp pool LAN2
Router(dhcp-config)#network 192.168.2.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.2.254
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
```

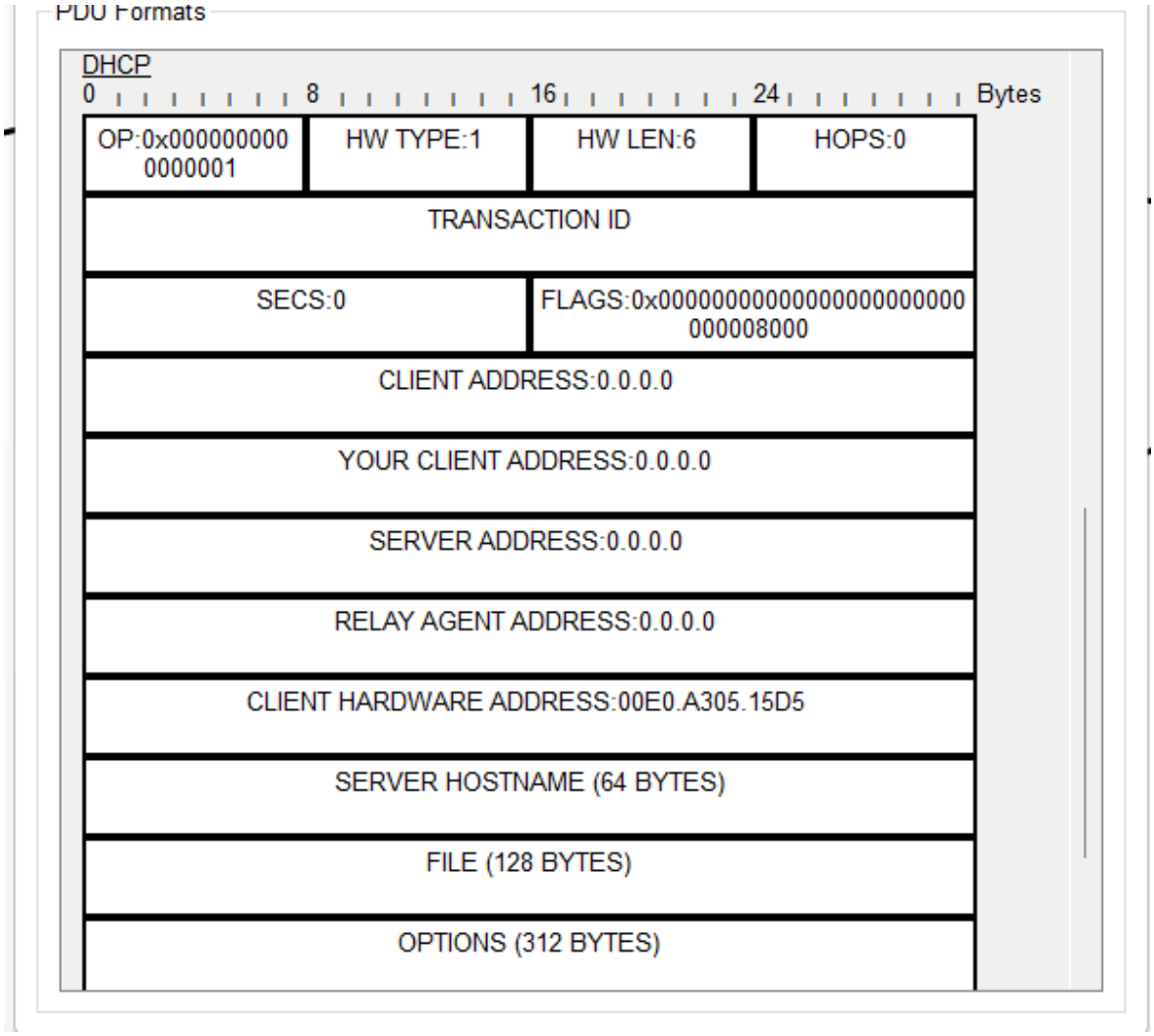


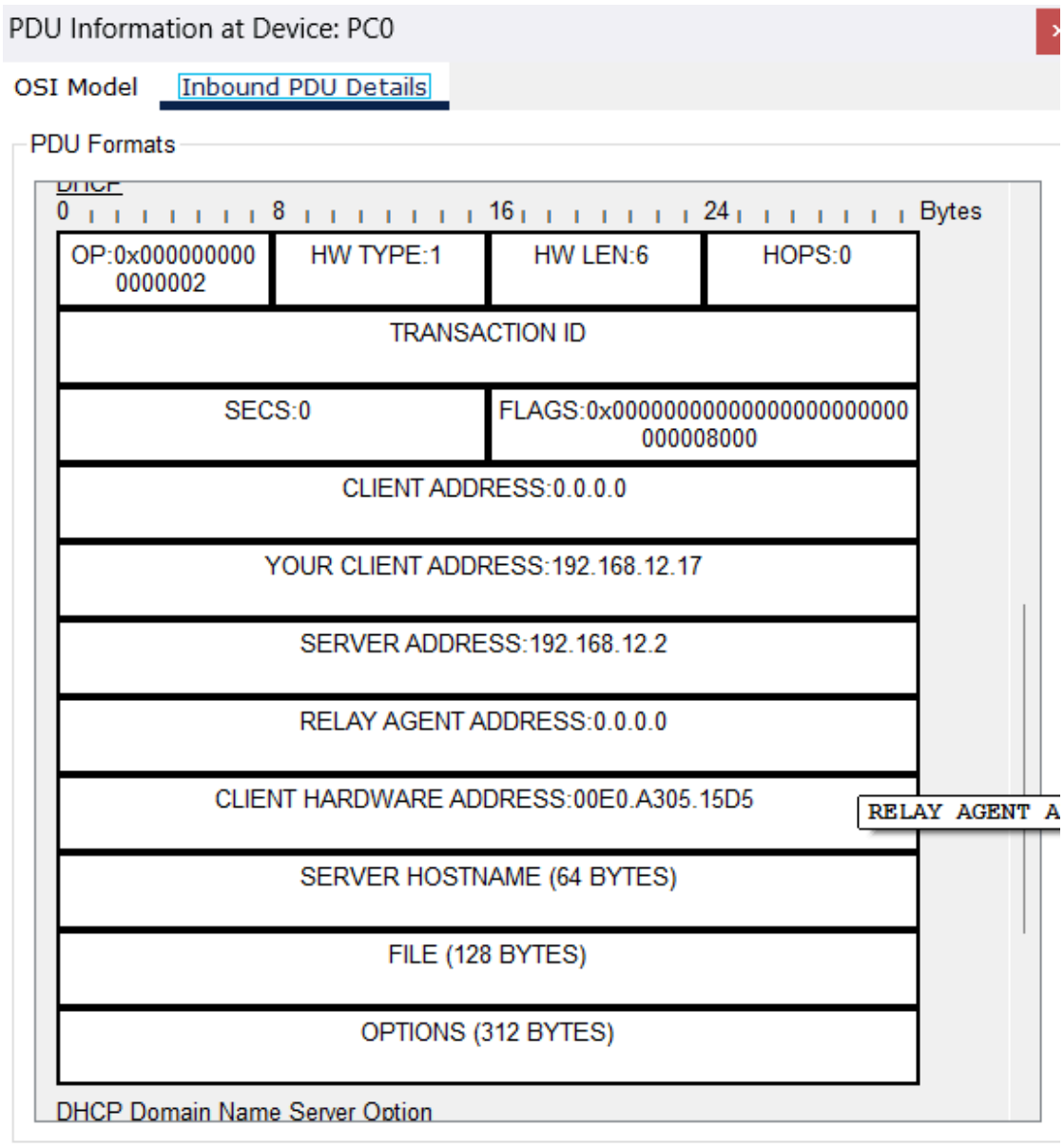


State diagram for server and client model of Socket

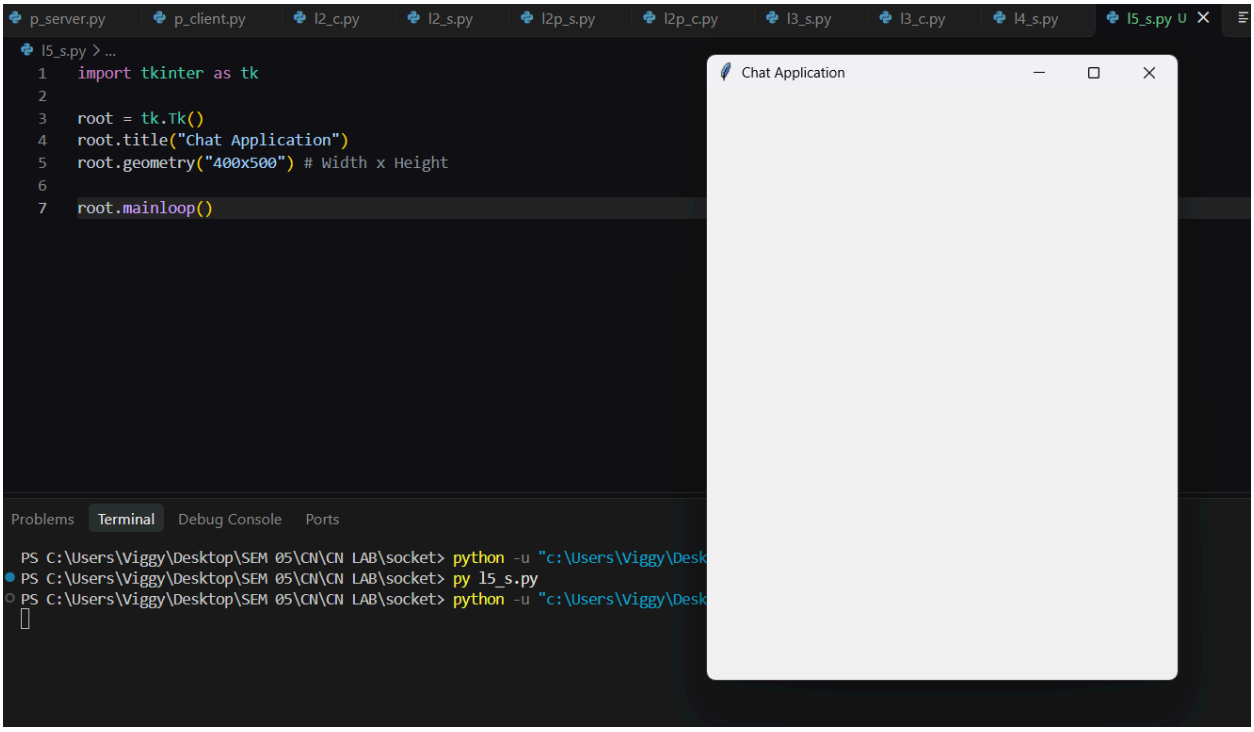
Function Call	Description
Socket()	To create a socket
Bind()	It's a socket identification like a telephone number to contact
Listen()	Ready to receive a connection
Connect()	Ready to act as a sender
Accept()	Confirmation, it is like accepting to receive a call from a sender
Write()	To send data
Read()	To receive data
Close()	To close a connection


MULTI-THREADED SOCKET SERVER ARCHITECTURE:
CONCURRENT WORD HANDLING
CLIENT 1
CLIENT 2
CLIENT N
Accept Client Connection (TCP)
MAIN ACCEPTOR THREAD
SPAWNS/ASSIGNS CLIENT HANDLING THREADS
THREAD 1
THREAD 2
THREAD N
CLIENT HANDLING THREAD WORKFLOW (e.g., THREAD i)
1. RECEIVE WORD FROM CLIENT
PARSE INPUT DATA
3. LOCK GLOBAL STRING MUTEX
(Acquire Write/Update Lock)
4. CHECK WORD EXISTENCE & APPEND IF NEW
apple orange banana mango...
If Word Present? → [Log Match]
If New? → [Append Word + Space, e.g., 'grape ']
6. GENERATE UPDATED GLOBAL STRING VIEW
7. QUEUE UPDATE FOR OTHER CLIENTS
(to Update Threads)
PREPARE "ACK/UPDATE" MESSAGE
8. SEND "ACK/UPDATE" BACK TO CLIENT
GLOBAL STRING ACCESS & SYNCHRONIZATION
GLOBAL STRING MUTEX
CLIENT HANDLING THREADS (1 to N)
ACQUIRE LOCK
GLOBAL STRING
WRITE LOCK (MUTEX)
SWRITE LOCK (Shared Word List ers)
SUCCESS
QUEUE/WAIT
THREAD 1
THREAD 2
THREAD N
READ ONLY ACCESS
(When Not Locked For Write)
GLOBAL STRING ACCESS & SYNCHRONIZATION
GLOBAL STRING MUTEX
CLIENT HANDLING THREADS (1 to N)
ACQUIRE LOCK
GLOBAL STRING
WRITE LOCK (MUTEX)
SUCCESS
THREAD 1
THREAD 2
THREAD 3
QUEUE/WAIT
THREAD N
THREAD ...
READ ONLY ACCESS
(When Not Locked For Write)
UDP/TCP UPDATE BROADCAST MECHANISM
UPDATE SENDER THREAD
BROADCAST TO CLIENTS
UPDATE: "apple orange"
UPDATE: "apple orange banana..."
UPDATE: "apple orange banana mango..."
CLIENT RECEIVES UPDATED WORD LIST


PDU Formats
DHCP
0
8
16
24
Bytes
OP:0x0000000000000001
HW TYPE:1
HW LEN:6
HOPS:0
TRANSACTION ID
SECS:0
FLAGS:0x00000000000000000000000000008000
CLIENT ADDRESS:0.0.0.0
YOUR CLIENT ADDRESS:0.0.0.0
SERVER ADDRESS:0.0.0.0
RELAY AGENT ADDRESS:0.0.0.0
CLIENT HARDWARE ADDRESS:00E0.A305.15D5
SERVER HOSTNAME (64 BYTES)
FILE (128 BYTES)
OPTIONS (312 BYTES)



Just A simple Example On How Do We Run



When u wanna give dhcp from router itself commands u need to give :

1	Convert lowercase text to uppercase	Use `data.upper()`
2	Convert text to lowercase	Use `data.lower()`
3	Reverse the received string	Use `data[::-1]`
4	Find the length of a string	Use `len(data)`
5	Count vowels in a string	Count `a, e, i, o, u`
6	Check whether a string is a palindrome	Compare with its reverse
7	Count words in a sentence	Use `len(data.split())`
8	Remove spaces from a string	Use `data.replace(" ", "")`
9	Count uppercase and lowercase characters	Check `isupper()` and `islower()`
10	Replace a character in a string	Use `data.replace()`